Practice Set Midterm I (not graded)

PHYS 6020 - Fall 2023

September 29, 2023

In addition to reading/practicing with the solved problems in the book, I'd suggest you try to work out through these to check your understanding.

Problem 1

Solve Exercise 2.25 at the end of Chapter 2 from Zettili's book.

Problem 2

Using results from previous HW sets calculate the matrices representing:

(i) $e^{2i\sigma_x}$

(ii) $(e^{i\sigma_x})^2$

(iii) $e^{i(\sigma_x+\sigma_y)}$

Is $e^{2i\sigma_x}$ equal to $(e^{i\sigma_x})^2$? Is $e^{i(\sigma_x+\sigma_y)} = e^{i\sigma_x}\, e^{i\sigma_y}$?

Problem 3

Solve Exercise 2.27 at the end of Chapter 2 from Zettili's book.

Problem 4

Solve Exercise 2.46 at the end of Chapter 2 from Zettili's book.